

BRUNO HENRIQUE FREITAS DE MELO

R&D SOFTWARE ENGINEER | BACKEND-HEAVY FULLSTACK & APPLIED AI

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PROFESSIONAL SUMMARY

R&D Software Engineer with a solid theoretical and practical background in backend systems development, software architecture, and applied artificial intelligence. B.S. in Information Systems (UFAC, Thesis Grade: 10.0/10.0), Postgraduate Specialization in Software Testing Engineering with AI (IARTES/Motorola, current GPA: 9.65/10.0), and M.Sc. candidate in Computer Science (PPGCC/UFAC). End-to-end software development lifecycle experience: building scalable RESTful APIs in Java (Spring Boot) and Python (Django REST Framework), high-performance relational database modeling (MySQL, PostgreSQL), and stateless JWT authentication with contextual role-based access control (RBAC). Rigorous software quality practices through Test-Driven Development (TDD with JUnit 5/Mockito), Behavior-Driven Development (BDD with Cucumber/Gherkin), static code analysis, and linters. Author of peer-reviewed paper published by the Brazilian Computer Society (SBC/BrasNAM 2025) featuring concurrent large-scale data mining (+136,000 records), alongside cutting-edge research in multi-agent refactoring systems (local LLMs via Ollama) and fuzzy inference systems (Mamdani). Practical experience as Scrum Master facilitating technical backlog refinement and continuous team alignment.

PROFESSIONAL EXPERIENCE

Web Academy – Motorola Mobility / UFAC Partnership

Jun 2025 – Dec 2025

Fullstack Developer & Scrum Master (Hands-On)

- Developed microservices and RESTful APIs in Java with Spring Boot, Maven, and JPA/Hibernate persistence, applying SOLID principles and Clean Architecture for high cohesion and maintainability.
- Modeled relational domain in MySQL with 6 interdependent entities, designing JPA queries with dynamic pagination and optimized filters for high request throughput.
- Implemented security layer with stateless JWT authentication and project-contextual role-based access control (dynamic RBAC) via Spring Security, isolating permissions by user scope.
- Standardized and documented over 30 RESTful endpoints via OpenAPI/Swagger with strict response contracts and uniform defensive exception handling (RFC 7807).
- Built responsive frontends in Angular and TypeScript, integrating HTTP interceptors, reactive forms, and synchronized session state management with backend.
- Diagnosed and resolved critical production incident regarding multipart upload for image backup to Google Drive via Controller-layer code inspection with zero downtime.
- Served as Scrum Master: led agile ceremonies (Dailies, Planning, Review, Retrospective), technical requirements refinement, and continuous stakeholder alignment.

National Social Security Institute (INSS)

Aug 2024 – Nov 2024

Backend Developer (Intern)

- Developed RESTful API with Django REST Framework for asset management and audit trails in public administration, modeling 9 relational entities with transactional integrity.
- Implemented JWT authentication, defensive exception handling, and standardized 100% of API response contracts with strict payload validation.
- Built bulk CSV export routines for seamless integration with public servant administrative systems.
- Autonomously drove end-to-end lifecycle, from functional requirements elicitation and validation with civil servants to final production sign-off.

PUBLICATIONS

Perceptions on Deforestation and Wildfires in Brazil: An Analysis on Bluesky

2025

Paper published at BrasNAM / CSBC – Brazilian Computer Society (SBC)

- Melo, B. H. F., et al. (2025). Published in the Proceedings of the XIV Brazilian Workshop on Social Network Analysis and Mining (sol.sbc.org.br).
- Engineered multithreaded Python pipeline with `ThreadPoolExecutor` for concurrent mining of +136,000 Bluesky posts via AT Protocol with exponential backoff retries and zero error rate.
- Applied lexical sentiment analysis with LeIA and topic modeling with LDA (*Latent Dirichlet Allocation*), correlating polarity with network engagement metrics. Foundation of Undergraduate Thesis (Grade 10.0/10.0).

SELECTED PROJECTS & ACADEMIC RESEARCH

RefAgent-Py: Code Refactoring with Multi-Agent Systems in Python

2026

Researcher / PPGCC/UFAC / Co-authored with Adrias Souza | github.com/I-Am-BrunoHFMelo/RefAgent-Python

- Research paper prepared for WES 2026 adapting the RefAgent multi-agent architecture (Planner, Generator, Compiler, Tester) to Python using native AST, mypy, and Pylint.
- Evaluated local open-weight models (Qwen, Llama) via Ollama on RTX 5080 GPU, achieving up to 62% improvement in functional test pass rates on RefactorBench and 17 real Flask framework files.

Fuzzy Inference System for Social Media Post Temperature 2026

Researcher (Sole Author) / M.Sc. PPGCC/UFAC / Python, scikit-fuzzy, NLP

- Technical paper proposing Mamdani Fuzzy Inference System (27 rules) to compute a Temperature Index ($T \in [0, 100]$), overcoming rigid hard-threshold limitations in NLP.
- Non-linearly fused Polarity (VADER), Subjectivity (TextBlob), and logarithmic-compressed Engagement (mitigating power laws), validated across 732 real Twitter/Kaggle posts.

BDD Library: Automated Testing with Cucumber & JUnit 5 2026

Java Developer / Postgraduate IARTES / github.com/I-Am-BrunoHFMelo/bdd-biblioteca-cucumber

- Modeled domain business rules for book loans and penalty calculations in Java with rigorous unit testing under JUnit 5.
- Implemented Behavior-Driven Development (BDD) acceptance tests with Cucumber and executable Gherkin specifications, establishing living documentation for the system.

Mini-CRM with Event-Driven Architecture & WebSockets 2026

Backend Developer / Clean Architecture / github.com/I-Am-BrunoHFMelo/laravel-mini-crm-contatos

- Event-driven architecture in PHP/Laravel with real-time WebSockets (Laravel Reverb) and asynchronous background queue processing with Redis.
- Applied Domain-Driven Design (DDD), Value Objects, Domain/App/Infra separation, and RFC 7807 error specifications.

Deck & Dice: Native Android App in Java (SDK 34) 2026

Mobile Developer / github.com/I-Am-BrunoHFMelo/Deck-Dice

- Developed native Android application in Java featuring interactive 2D Canvas with physics, custom dice expression evaluation engine, and local relational persistence with Room Database (SQLite).
- Applied advanced object-oriented design, Java collections, and Android platform lifecycle management.

Anima Ultima: Offline-First Mobile Companion App 2026

Frontend / Mobile Developer / github.com/I-Am-BrunoHFMelo/Anima-Ultima-Aplicacao-Mobile-para-Fabula-Ultima

- Responsive mobile companion app built with Ionic 8, Angular 20, and Capacitor for RPG character sheet management with retro 16-bit aesthetics.
- Offline-first local persistence using IndexedDB via Dexie.js with debounced auto-save routines and reactive state management with RxJS.

Single-Layer Perceptron Classifier in Python/NumPy 2026

Researcher / M.Sc. PPGCC/UFAC / github.com/I-Am-BrunoHFMelo/py-perceptron

- Vectorized implementation of Rosenblatt's Perceptron neuron in pure Python with NumPy, utilizing a hard-limiter activation function for linearly separable binary classification.
- Verified exact convergence of separation hyperplanes and analyzed quadratic error convergence on epoch 2 of training.

Personal Jamstack Portfolio in Astro 5 & Interactive Islands 2026

Frontend Developer / github.com/I-Am-BrunoHFMelo/BHFM-Portifolio-XXV

- Engineering hub developed with Astro 5, TypeScript, Tailwind CSS, and Islands Architecture, achieving top scores on Google Lighthouse.
- Integrated Web Audio API for interactive synthesized positional audio directly in the browser with automated CI/CD deployment on Vercel.

Operations Research: 0/1 Knapsack Optimization Algorithm 2025

Researcher / B.S. UFAC / github.com/I-Am-BrunoHFMelo/knapsack-PesquisaOperacional_2025

- Mathematical modeling and Python implementation of linear optimization algorithms for the 0/1 Knapsack problem, comparing greedy algorithms and dynamic programming.
- Asymptotic time and space complexity analysis across large problem instances with convergence benchmarking.

ACADEMIC ACTIVITIES, EXTENSION & LEADERSHIP

Accessibility Support Center (NAI / UFAC) 2022 – 2025

Assistive Technology & Accessibility Fellow (1,360 hours)

- Provided technical accompaniment and pedagogical assistance to student with motor disability throughout undergraduate degree, internship, and final thesis defense.
- Supported maintenance of institutional web portal for PPGCC/UFAC and Arduino prototyping for academic Wi-Fi automation research.

Organizing Committee of University Extension Project 2023

Student Collaborator – Agile Project Management (Jira, Trello, Asana, Teamwork, RunRun.it)

- Organized and led training workshops for the academic community on agile workflows and project management software.

III Northern Competitive Programming Marathon 2023

Technical Support & Organizing Volunteer

- Provided logistical and operational support for regional competitive programming contest hosted at Federal University of Acre.

- Participated in panel discussions on advancements in Artificial Intelligence, Software Systems, and Computer Networks.

EDUCATION & SPECIALIZED TRAINING

M.Sc. in Computer Science (PPGCC) <i>Federal University of Acre (UFAC) – Research: Artificial Intelligence & Software Systems</i>	2026 – 2028 (In Progress)
Postgraduate Specialization in Software Testing Engineering with AI (IARTES) <i>Federal University of Acre in partnership with Motorola Mobility (Current GPA: 9.65/10.0)</i>	2026 – 2027 (In Progress)
Bachelor of Science in Information Systems <i>Federal University of Acre (UFAC) – Final Thesis Grade: 10.0/10.0</i>	2019 – 2025 (Completed)
Web Academy: Intensive Fullstack Development Program (300h) <i>Federal University of Acre in partnership with Motorola Mobility (Java, Spring Boot, Angular, Scrum)</i>	2025 (Completed)
N.A.V.E. Tech Acre: Innovation, Technology & Entrepreneurship <i>Federal University of Acre in partnership with Samsung (Lean Startup, Canvas, Figma, Pitch)</i>	2022 (Completed)

TECHNICAL SKILLS

Languages & Backend: Java (17/21), Python, PHP, TypeScript, JavaScript, SQL. Spring Boot, Spring Security (JWT, RBAC), JPA/Hibernate, Maven, Django REST Framework, Laravel, Reverb (WebSockets), SOLID Principles, Clean Architecture, RESTful APIs, OpenAPI/Swagger, RFC 7807, Distributed Systems fundamentals.

Artificial Intelligence & Data Engineering: Multi-Agent Systems (local LLMs via Ollama), Natural Language Processing (NLP/spaCy, NLTK, VADER, TextBlob, Gensim/LDA), Fuzzy Logic (scikit-fuzzy, Mamdani), Fundamental Neural Networks (Perceptron/NumPy), Concurrent Programming (`ThreadPoolExecutor`, exponential backoff), ETL & data mining.

Software Quality & Testing: Test-Driven Development (TDD), JUnit 5, Mockito, BDD with Cucumber (executable Gherkin specifications), PyTest, Static Code Analysis (mypy, Pylint, AST), Unit & Integration Testing, Code Review.

Frontend, Mobile & Databases: Angular (19/20), RxJS, Ionic 8, Capacitor, Tailwind CSS, HTML5, CSS3, Native Android (SDK 34, Java, 2D Canvas), Astro 5 (Islands Architecture). MySQL, PostgreSQL, SQLite, Room Database, Redis, IndexedDB (Dexie.js).

DevOps, Methodologies & Tooling: Docker, Git/GitHub (Git Flow, Pull Requests), CI/CD, GitHub Actions, Linux. Agile Methodologies: Scrum (Scrum Master experience), Kanban, Jira, Trello, Asana, Teamwork, RunRun.it, Lean Startup, Design Thinking, Requirements Engineering.

LANGUAGES

Portuguese: Native.

English: Intermediate (B1/B2) – Advanced technical reading of documentation and scientific literature; clear technical writing for code, specifications, PRs, and incident documentation; auditory comprehension of technical subjects; active conversational practice.